provide a data analysis project which will highlights these decision making model laya's performance.suggest some examples.

A strong project is an **automated customer-support triage and risk dashboard**. It showcases Laya's three native decision types—`choice`, `score`, and `noul`—on realistic text data, and lets you measure accuracy, latency, confidence calibration, and business impact against a baseline. [1] [2]

Recommended project: Support Triage Intelligence

Goal: Automatically analyze incoming support tickets, classify them, prioritize them, detect churn/security risk, and route them to the right team—then compare Laya with a simpler baseline.

Laya is especially appropriate here because it accepts a ticket or JSON state and returns typed decisions with probabilities in a single forward pass, rather than generating free-form text. [3] [1]

Data

Use a public support-ticket dataset, such as a Kaggle support-ticket classification set with 2,000 tickets across five issue categories. [4]

For each ticket, create labels for:

- **Category:** billing, account access, technical issue, product request, cancellation.
- **Priority:** low, medium, high, urgent.
- **Escalation risk:** yes/no.
- **Churn-risk score:** 0–10.
- **Routing team:** Billing, Technical Support, Account Management, Security, Retention.

If the dataset lacks some labels, create a small "golden set" of 200–500 manually labeled examples, then use LLM-assisted labeling only as a draft with human verification.

Laya decision design

Analysis question	Laya type	Example prompt
What is the ticket category?	`choice`	"Classify this ticket as one of: billing, account_access, technical_issue, product_request, cancellation."
How urgent is it?	`score`	"Rate urgency from 1 to 5, where 5 means immediate business impact."

Analysis question	Laya type	Example prompt
Should it be escalated?	`noul`	“Does this ticket require escalation to a senior agent?”
Is there churn risk?	`noul`	“Does the customer explicitly threaten cancellation or express intent to leave?”
Which team should receive it?	`choice`	“Route this ticket to: Billing, Technical Support, Account Management, Security, or Retention.”
Is this a security concern?	`noul`	“Does this message indicate account compromise, phishing, or unauthorized access?”

This directly exercises Laya’s documented capabilities: typed choices, ordered scores, and calibrated yes/no probabilities. [5] [1]

Metrics to highlight

Metric	What it shows	Why it matters
Accuracy / macro-F1	Correct category and routing decisions	Measures core classification quality
Mean absolute error	Difference between predicted and true priority/churn score	Evaluates score decisions
ROC-AUC / PR-AUC	Ability to rank escalation, churn, and security risk	More informative than raw accuracy for rare events
Calibration error	Whether a stated 80% probability is right about 80% of the time	Tests Laya’s probability quality
Latency	Milliseconds per decision	Highlights its fast, single-pass design
Throughput	Tickets processed per second	Shows suitability for high-volume automation
Cost per 1,000 tickets	Local or API cost	Demonstrates operational advantage
Human-review reduction	Percentage of tickets auto-routed without manual triage	Translates model performance into business value

Laya’s published typed-decisions benchmark reports 76.6% accuracy across 2,000 decisions for its fine-tuned checkpoint, versus 36.2% for its base English checkpoint, so your project should include a fine-tuned or task-adapted configuration if possible. [2] [6]

Baselines to compare

- **Keyword rules:** “refund,” “urgent,” “cancel,” “password,” “fraud.”
- **Majority-class baseline:** always predict the most common category or “do not escalate.”
- **Classical ML:** TF-IDF plus logistic regression or a gradient-boosted classifier.
- **General-purpose LLM:** a larger generative model with JSON output, compared on accuracy, latency, and cost.
- **Fine-tuned Laya:** ideally trained or adapted on your labeled ticket set.

This comparison is important because Laya's specialized checkpoints can substantially outperform its general checkpoint, while general LLMs may be stronger for nuanced reasoning but slower and more expensive. [6] [7]

Example outputs

Input ticket:

> "My payment failed twice this month and I still cannot access the premium dashboard. If this is not fixed today, I will cancel our company plan."

Expected Laya decisions:

- Category: `billing` or `technical_issue`
- Urgency: 5/5
- Escalation: `true`, high probability
- Churn risk: `true`, high probability
- Route to: `Retention` or `Billing`
- Security concern: `false`

Dashboard output:

- 82% of tickets auto-classified correctly
- 91% of urgent tickets correctly prioritized
- 76% of tickets eligible for no-touch routing
- Median decision latency: about 33 ms per ticket
- Estimated analyst time saved: 6 hours per 1,000 tickets

The roughly 33 ms single-pass decision time is a headline performance metric worth reporting, though your measured latency will depend on hardware and deployment. [2]

Optional extensions

- **Phishing and email triage:** classify emails as legitimate, spam, phishing, or account-security risk; Laya's research workflows explicitly include support triage, email/phishing, guardrails, RAG relevance, moderation, and model routing. [8]
- **Review analytics:** classify product reviews by topic, score satisfaction, and flag churn or safety concerns.
- **RAG relevance filter:** decide whether retrieved documents actually answer a user query before sending them to a chatbot. [8]
- **Content moderation:** classify user-generated content and estimate policy-violation probability. [8]
- **Multilingual test:** run the same ticket triage workflow in English plus one or more additional languages to demonstrate Laya's multilingual design. [2]

Deliverables

1. Cleaned and labeled ticket dataset.
2. Laya decision schema for category, urgency, escalation, churn, security, and routing.
3. Baseline models and evaluation notebook.
4. Confusion matrices, calibration plots, latency distribution, and cost analysis.
5. Streamlit, Dash, or Power BI dashboard showing ticket volume, risk queue, routing distribution, and automation rate.
6. Final report recommending which decisions can safely be automated and which require human review.

This project is compelling because it does not merely test whether Laya can classify text—it demonstrates an end-to-end analytics workflow where its fast, structured decisions reduce manual triage, surface risk earlier, and improve operational efficiency. [7] [9]

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1. https://github.com/receptron/laya
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15. https://github.com/NandhaKishorM/laya